

prospective

AI-assisted architecture

workload intent → evidence-backed finding or decision

2010s–2020s

Agile hardware generation

parameterized templates → bespoke SoC instances

2000s–2010s

Domain-specific synthesis

specification + DSL intent → executable code

1990s–2000s

High-level synthesis

algorithmic behavior → RTL candidates

1980s–1990s

Logic synthesis

Boolean and RTL intent → gate-level structure

time (era spans approximate)

widening unit of
intent and artifact